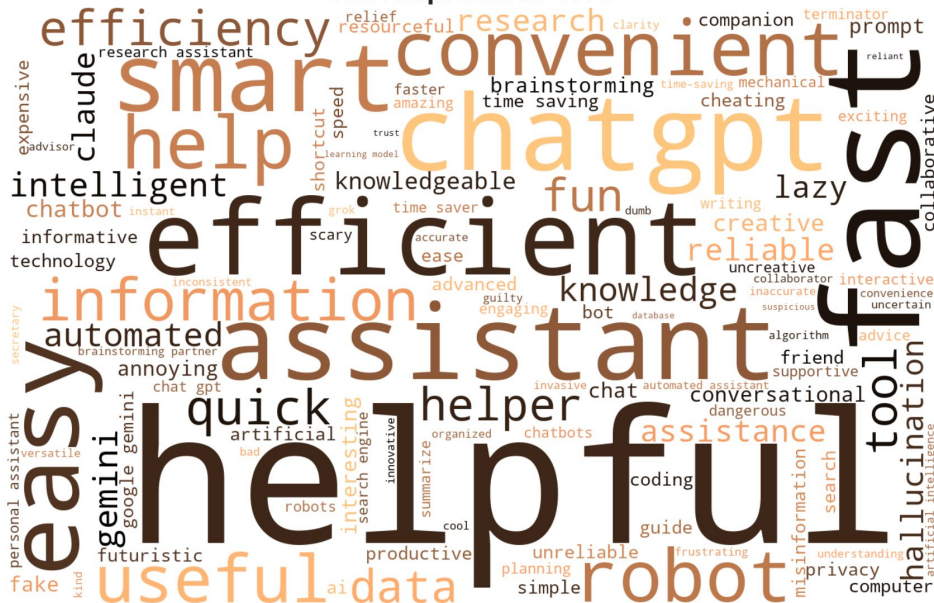


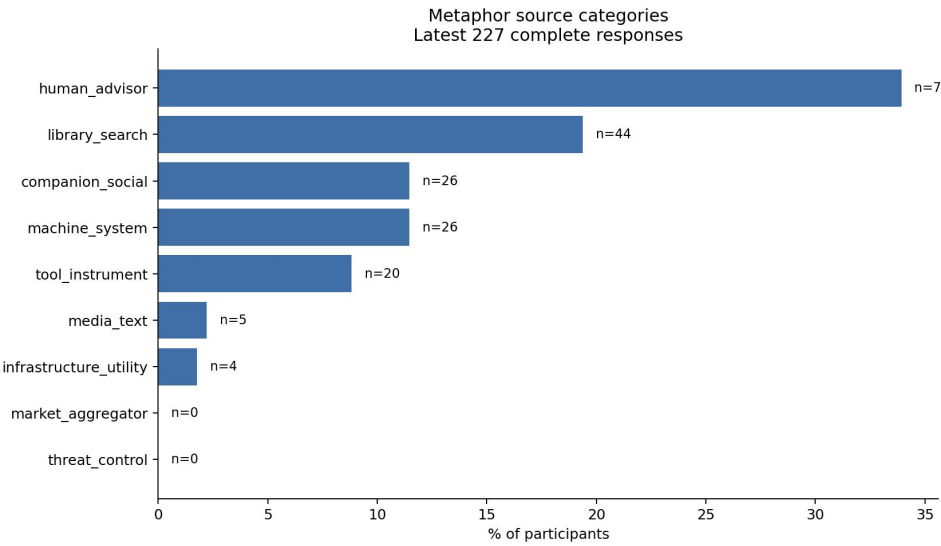
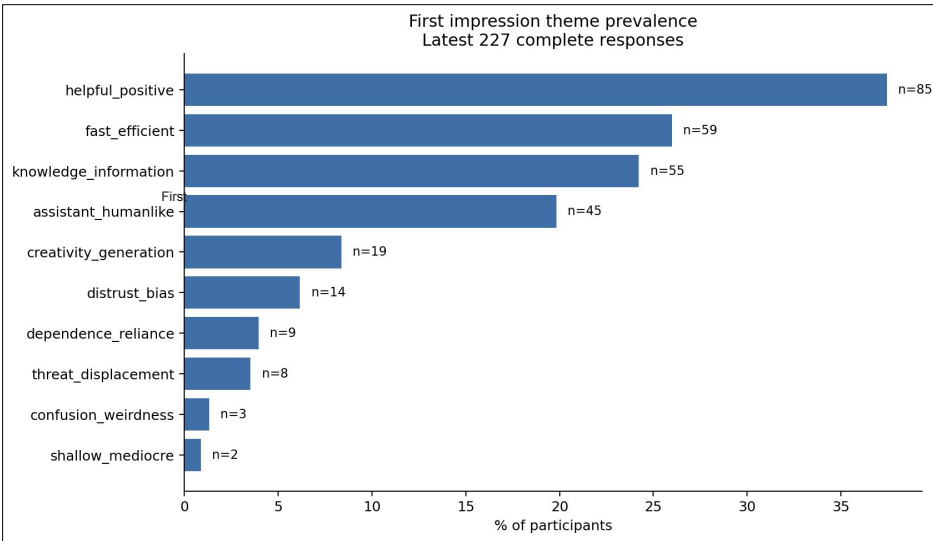
# People's Mental Model & Behavior Usage of AI

one analysis/figure per slide

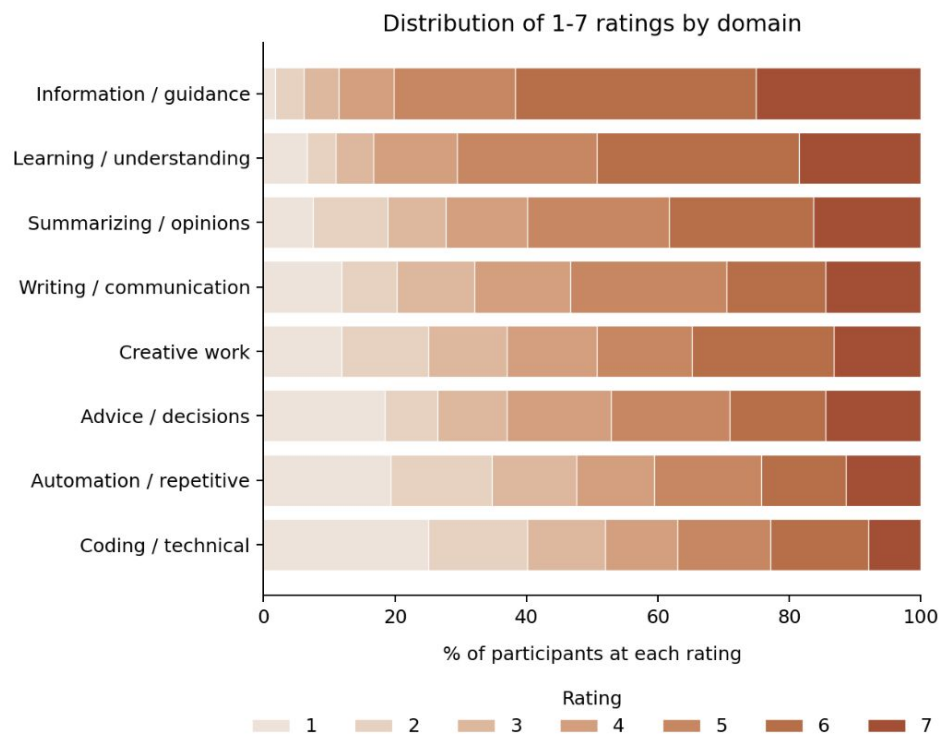
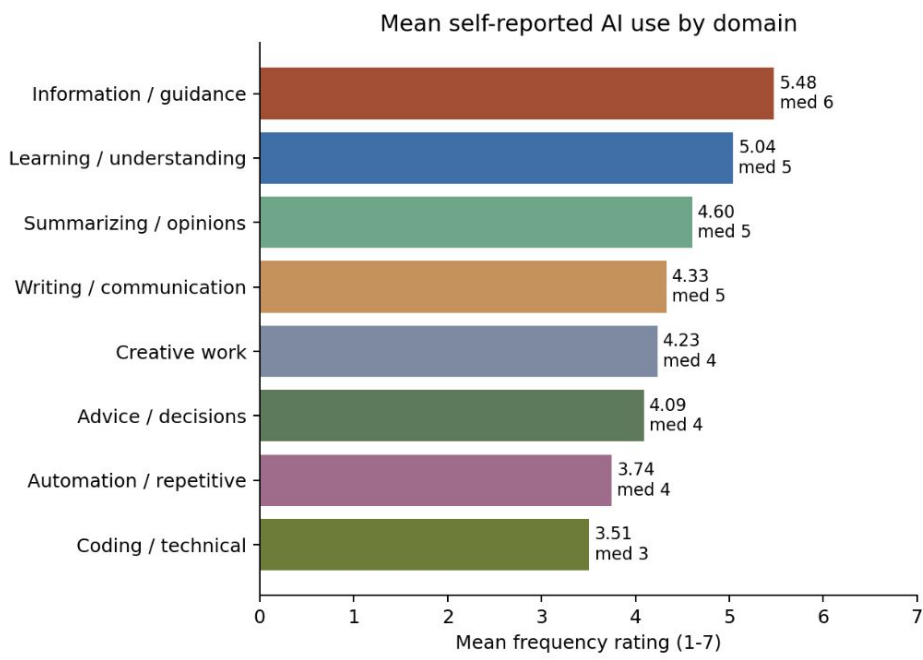
# People's first impressions & metaphors of AI



# People's first impressions & metaphors by theme prevalence

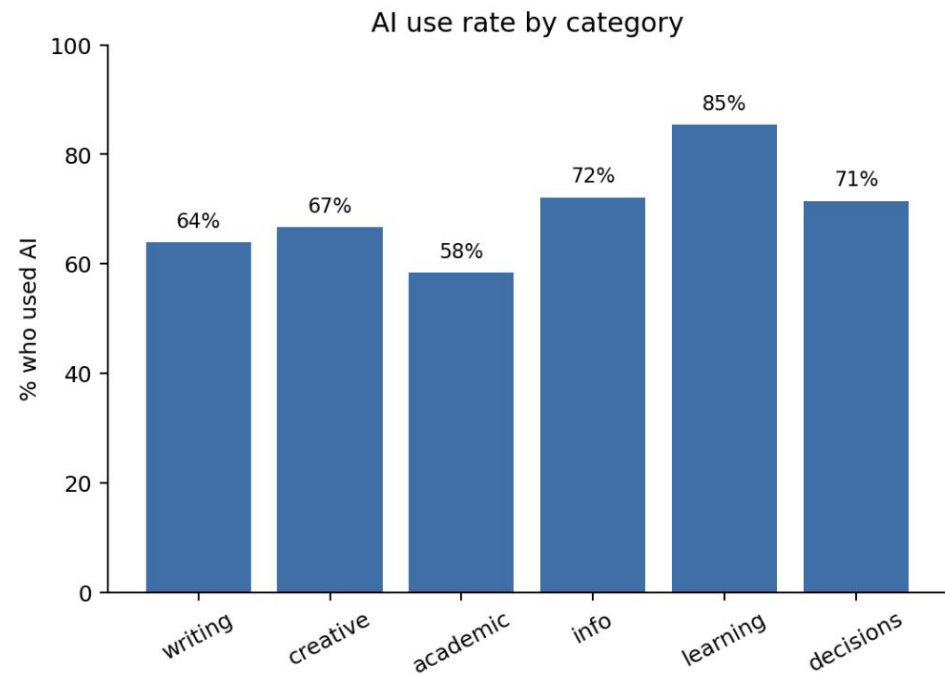
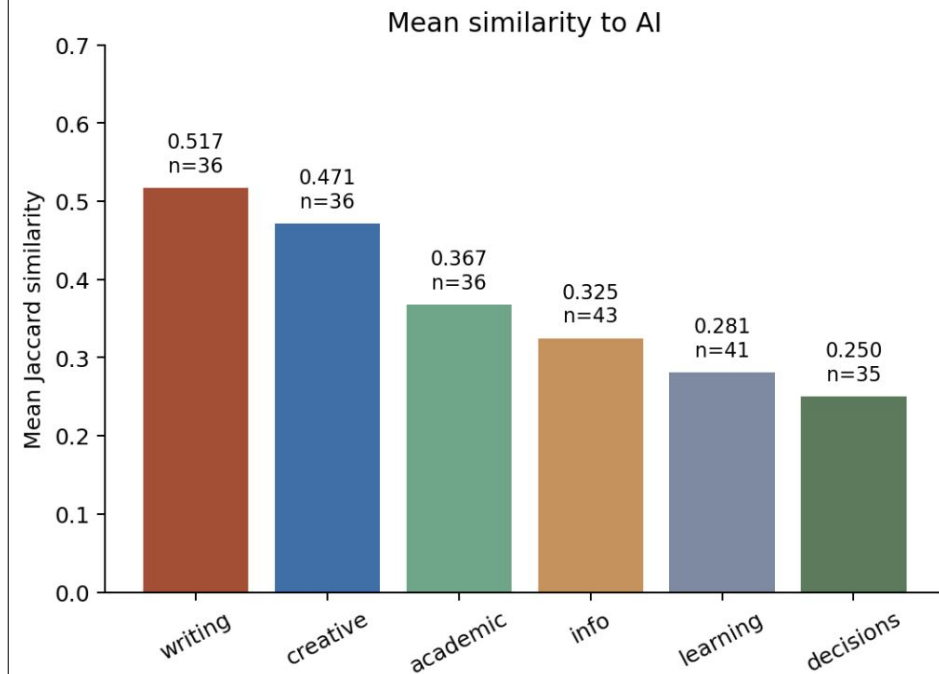


# How much people report using AI by domain

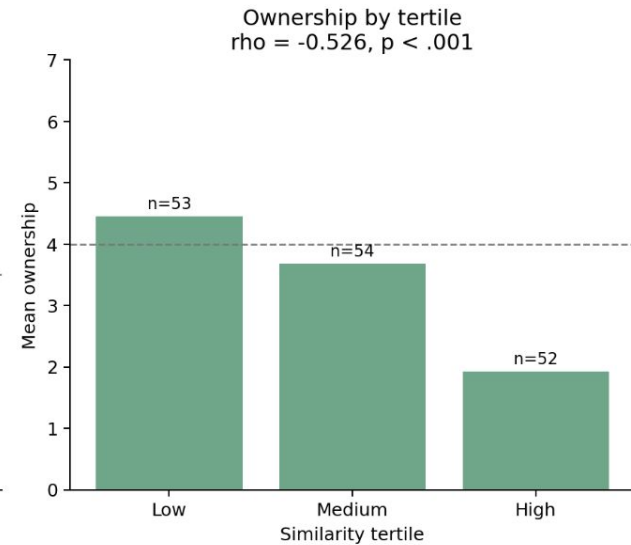
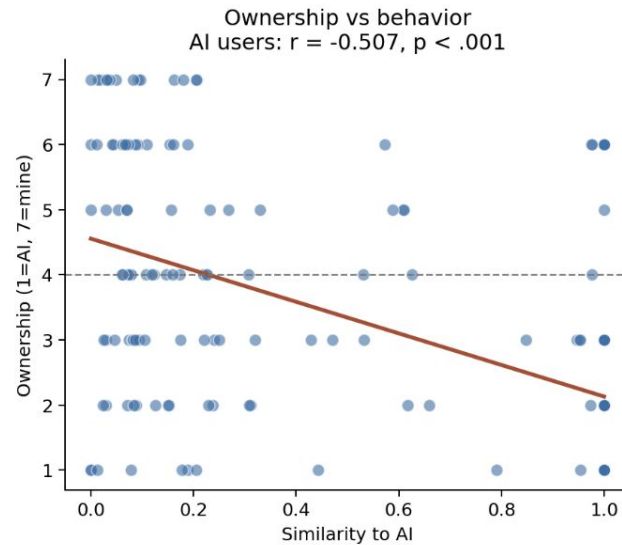
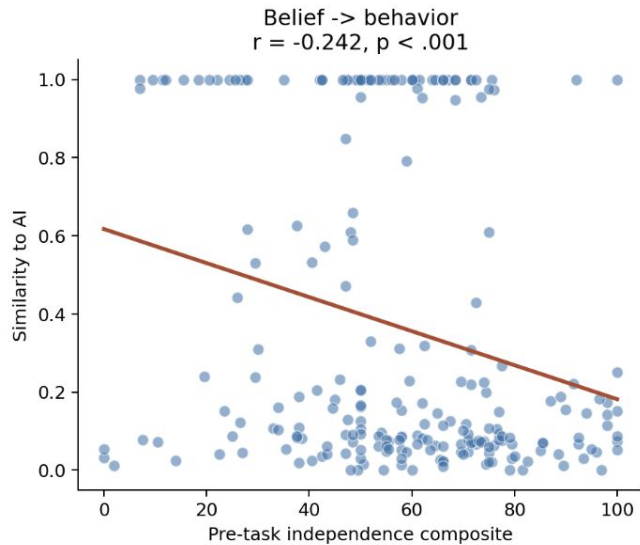


# Mean Jaccard Similarity to AI & AI Use by Category

## Latest 227 complete responses



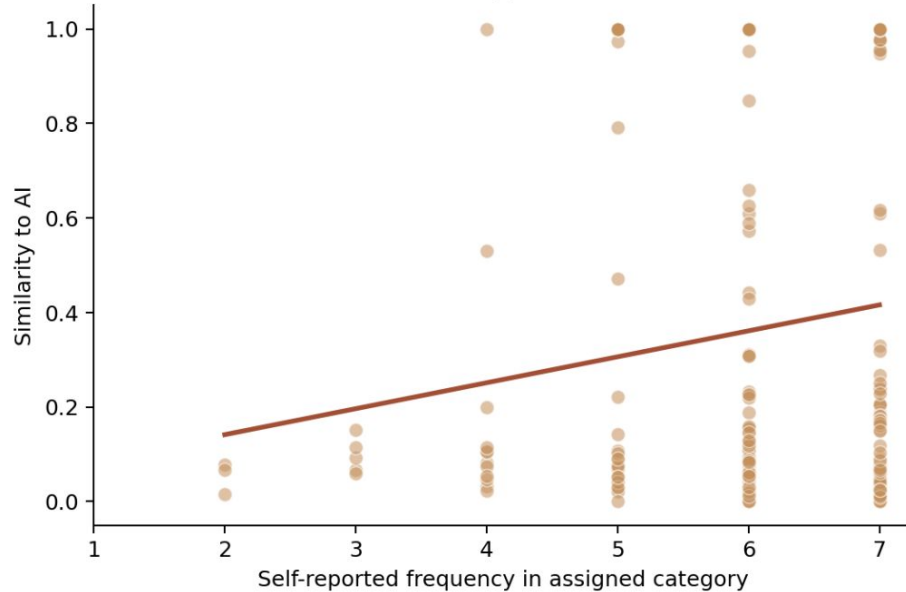
In general, people respond less similarly to AI when they report being more independent and report less epistemic ownership when their response is more similar to AI



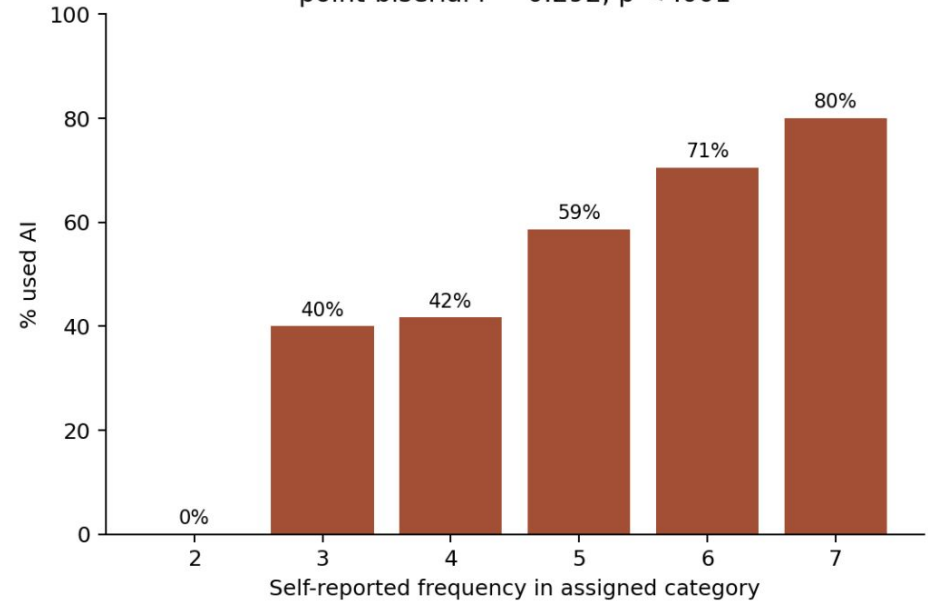
# People who report using AI for the assigned task category respond more like AI

## Self-reported use and behavior

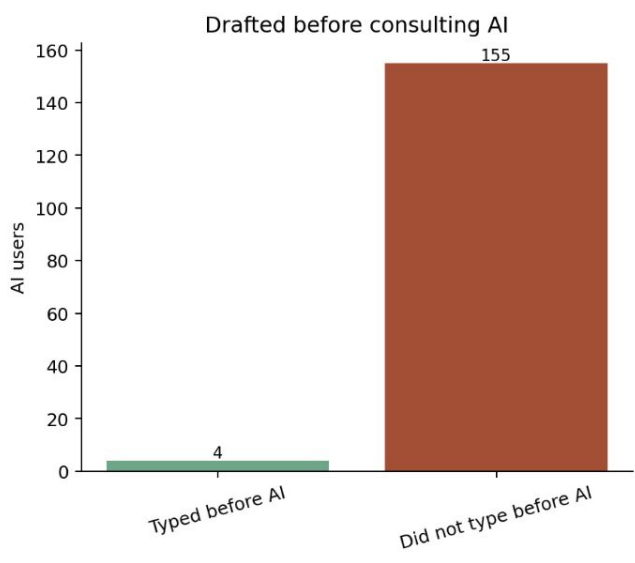
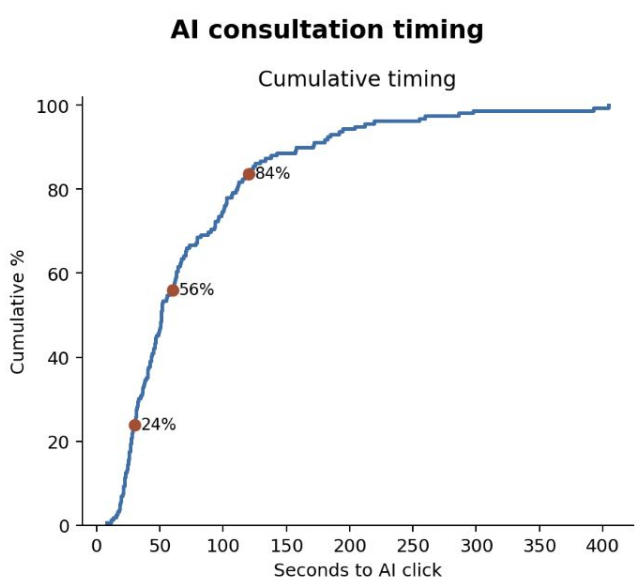
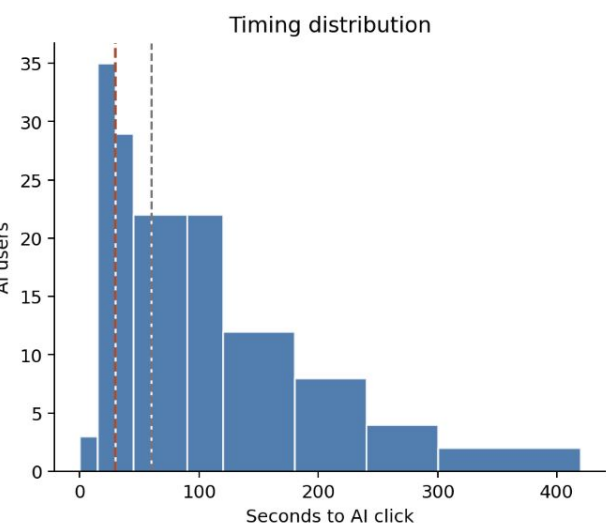
Frequency -> similarity  
 $r = 0.149$ ,  $p = 0.025$



Frequency -> AI use  
point-biserial  $r = 0.292$ ,  $p < .001$

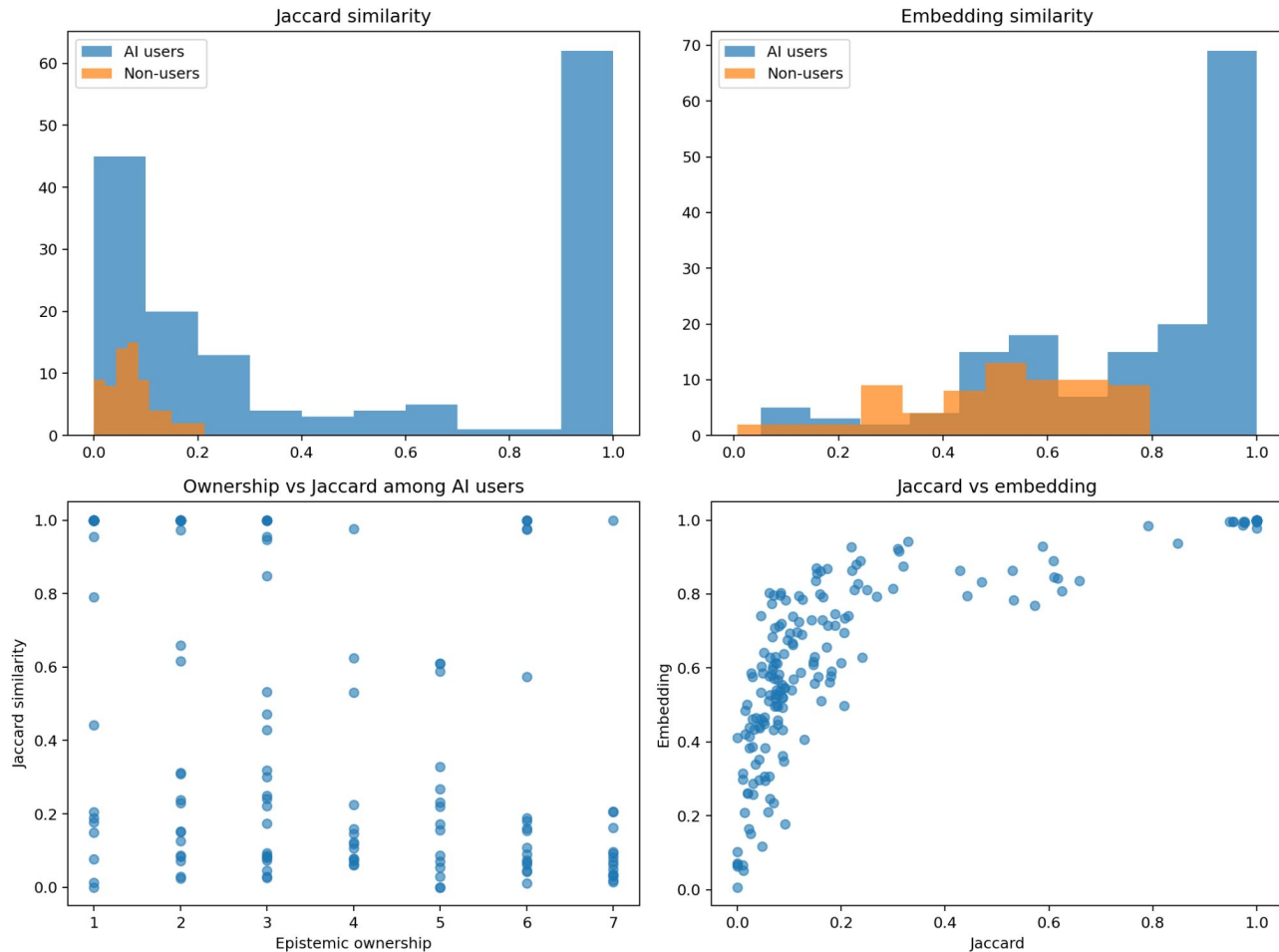


# Many people consulted AI very quickly.



# Jaccard Similarity vs Embedding Similarity

- Non-AI users show much lower Jaccard similarity (embedding similarity is closer to AI users).
- People who copy-paste (Jaccard-similarity == 1) still report some epistemic ownership.



```

## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: ai_used ~ z_user_decides + z_independent + z_hours_per_week +
##          z_instrument_vs_point_of_view + (1 | task_id)
## Data: model_data
## Control: glmerControl(optimizer = "bobyqa")
##
##           AIC          BIC      logLik -2*log(L)  df.resid
##        262.0        282.6     -125.0     250.0       221
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.7115 -0.9226  0.4116  0.7047  1.3366
##
## Random effects:
##   Groups Name            Variance Std.Dev.
##   task_id (Intercept) 0          0
## Number of obs: 227, groups: task_id, 18
##
## Fixed effects:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.9816    0.1660   5.912 3.38e-09 ***
## z_user_decides    -0.7946    0.1989  -3.996 6.45e-05 ***
## z_independent     -0.1246    0.1600  -0.779  0.4361
## z_hours_per_week    0.3117    0.1807   1.725  0.0846 .
## z_instrument_vs_point_of_view 0.1682    0.1591   1.057  0.2905
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) z_sr_d z_ndpn z_hr__
## z_user_dcds -0.350
## z_indepndnt -0.025 -0.166
## z_hrs_pr_wk  0.119 -0.053  0.032
## z_nstrm____  0.014 -0.020  0.133 -0.224
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

```

Participants who more strongly said the user does the work / makes the decisions were significantly less likely to consult AI,  $p < .001$ .

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## jaccard_similarity_to_ai ~ z_user_decides + z_independent + z_hours_per_week +
##   z_instrument_vs_point_of_view + (1 | task_id)
## Data: ai_users
##
## REML criterion at convergence: 191.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.6370 -0.8175 -0.2433  0.9534  1.5981
##
## Random effects:
##   Groups Name      Variance Std.Dev.
## task_id (Intercept) 0.02084  0.1444
## Residual            0.15934  0.3992
## Number of obs: 158, groups: task_id, 18
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    0.50358    0.04748  16.07656  10.607 1.15e-08
## z_user_decides -0.04194    0.03304  151.16617  -1.269   0.206
## z_independent  -0.04106    0.03465  150.52019  -1.185   0.238
## z_hours_per_week 0.01427    0.03207  146.94145   0.445   0.657
## z_instrument_vs_point_of_view 0.01494    0.03418  150.61159   0.437   0.663
##
## (Intercept)      ***
##
## z_user_decides
## z_independent
## z_hours_per_week
## z_instrument_vs_point_of_view
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) z_sr_d z_ndpn z_hr__
## z_user_dcds  0.126
## z_indepndnt  0.041 -0.213
## z_hrs_pr_wk -0.079 -0.087 -0.067
## z_nstrm____ -0.058 -0.177  0.203 -0.124
```

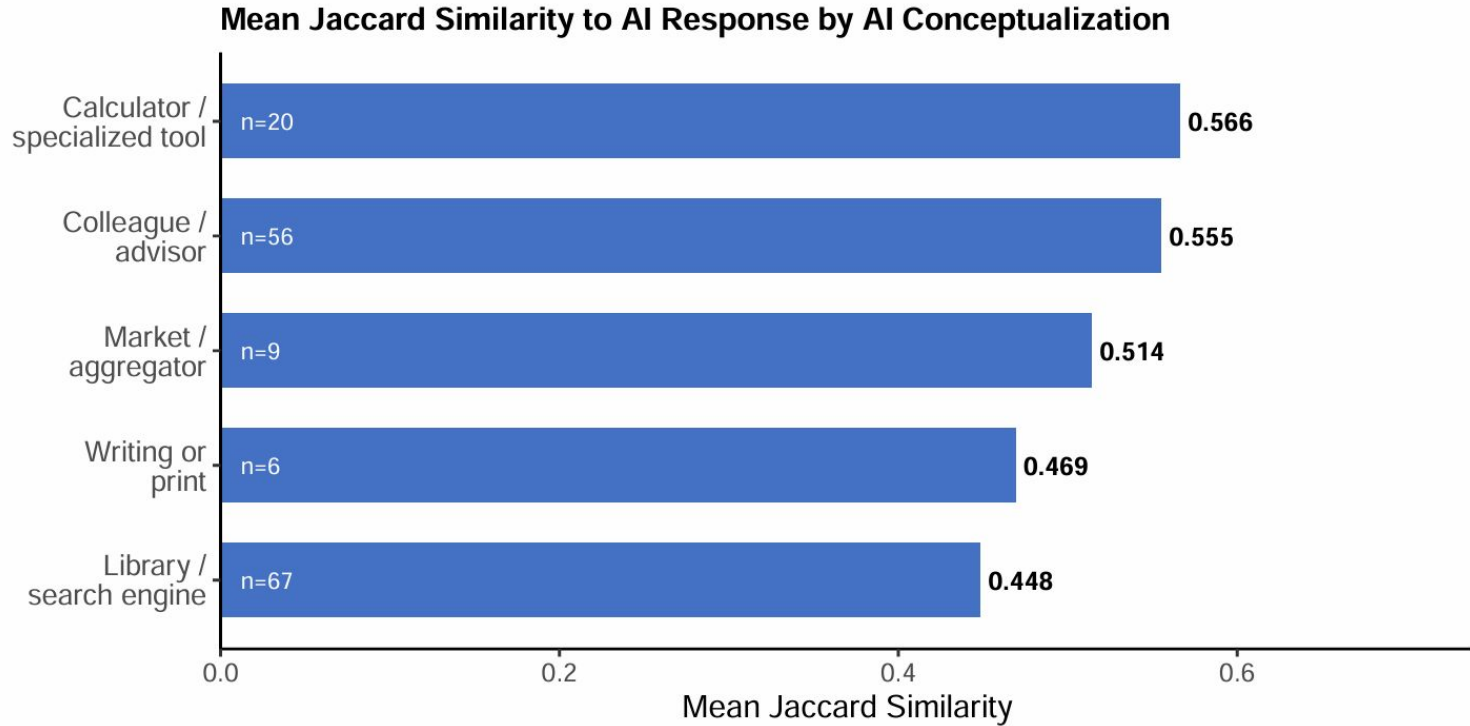
Among participants who used the AI assistant, neither pre-task independence predictor significantly predicted Jaccard similarity to the AI response:  $z\_user\_decides$ ,  $p = .206$ ;  $z\_independent$ ,  $p = .238$ .

```

## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula:
## epistemic_ownership_binary ~ z_jaccard_similarity_to_ai + z_user_decides +
## z_independent + z_hours_per_week + (1 | task_id)
## Data: ai_users
## Control: glmerControl(optimizer = "bobyqa")
##
##          AIC          BIC      logLik -2*log(L)  df.resid
##        96.1         109.9      -42.1      84.1         67
##
## Scaled residuals:
## -2.2523 -0.6975 -0.4001  0.7600  2.0327
##
## Random effects:
## Groups Name          Variance Std.Dev.
## task_id (Intercept) 0.2145   0.4631
## Number of obs: 73, groups: task_id, 18
##
## Fixed effects:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.4059    0.2989  -1.358  0.17452
## z_jaccard_similarity_to_ai -0.8932    0.3299  -2.708  0.00677 **
## z_user_decides    0.3446    0.2867   1.202  0.22946
## z_independent    -0.3240    0.2949  -1.099  0.27183
## z_hours_per_week    0.2307    0.2639   0.874  0.38216
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) z_j___ z_sr_d z_ndpn
## z_jccrd_s__ -0.046
## z_user_dcds  0.006 -0.077
## z_indepndnt  0.070  0.102 -0.175
## z_hrs_pr_wk -0.101 -0.177 -0.119 -0.012

```

Greater Jaccard similarity to the AI response significantly predicted a reduced likelihood of reporting primarily own-thinking ownership,  $p < .001$ .



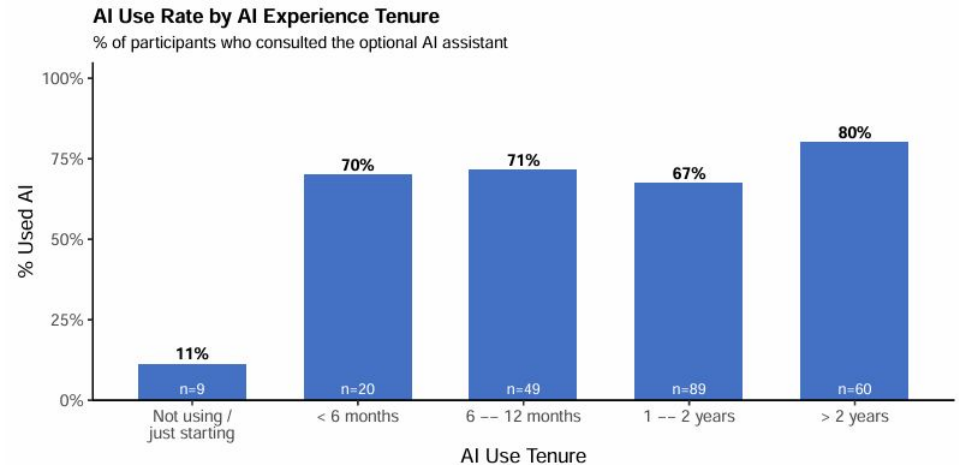
AI conceptualization did not significantly predict Jaccard similarity. Regardless of whether participants saw AI as a tool or a colleague, they copied its output at similar rates.

```
## Model:
## ai_used ~ ai_use_tenure + z_hours_per_week + ai_understanding +
## (1 | task_id)
##               npar    AIC      LRT Pr(Chi)
## <none>                279.80
## ai_use_tenure         4 285.74 13.9376 0.007497 **
## z_hours_per_week      1 278.89  1.0930 0.295815
## ai_understanding      3 274.48  0.6795 0.878007
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## ai_use_tenure               prob      SE df asymp.LCL
## I do not use them, or am just starting out. 0.145 0.1350 Inf  0.0196
## Less than 6 months                0.741 0.1050 Inf  0.4945
## 6 to 12 months                    0.747 0.0743 Inf  0.5774
## 1 to 2 years                      0.708 0.0656 Inf  0.5657
## More than 2 years                 0.813 0.0582 Inf  0.6722
## asymp.UCL
##      0.590
##      0.894
##      0.865
##      0.819
##      0.902
##
## Results are averaged over the levels of: ai_understanding
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

Participants with any established AI experience were similarly likely to consult the optional AI assistant, regardless of whether they had used AI for a few months or several years

AI use tenure significantly predicted use of the optional AI assistant,  $p = .0075$ . Participants who reported not using AI or just starting out had a much lower estimated AI use probability (.15) than all other tenure groups (.71 - .81)

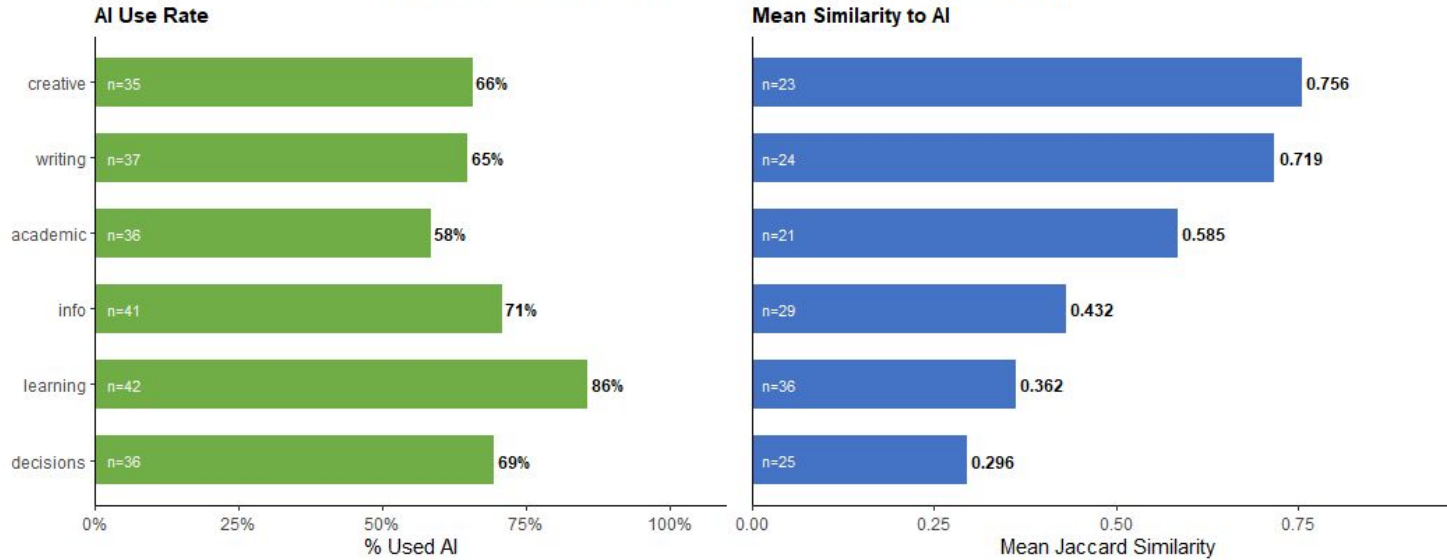


```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq  Mean Sq NumDF  DenDF F value Pr(>F)
## ai_use_tenure    0.68531 0.171328     4 143.31  1.0531 0.3821
## z_hours_per_week 0.00276 0.002759     1 143.19  0.0170 0.8966
## ai_understanding 0.27510 0.091700     3 142.26  0.5637 0.6398

## ai_use_tenure           emmean      SE    df lower.CL
## I do not use them, or am just starting out.  0.119 0.4220 146.4  -0.714
## Less than 6 months                        0.527 0.1230 139.6   0.283
## 6 to 12 months                          0.630 0.0863  98.3   0.459
## 1 to 2 years                             0.493 0.0734  70.4   0.347
## More than 2 years                        0.600 0.0776  80.8   0.446
## upper.CL
##      0.952
##      0.770
##      0.801
##      0.639
##      0.755
##
## Results are averaged over the levels of: ai_understanding
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95
```

AI use tenure did not significantly predict Jaccard similarity to the AI response,  $p = 0.382$ . Observed mean Jaccard ranged from .49 (1-2 years) to .63 across tenure groups, with no consistent trend with use tenure level.

AI Use Rate & Mean Jaccard Similarity to AI by Task Category



Among AI users, assigned task category significantly predicted Jaccard similarity,  $p < .001$ . Creative and writing tasks showed the highest similarity to the AI response, while decisions and learning showed the lowest. Tukey-adjusted comparisons showed that creative tasks were significantly higher than decisions ( $p = .001$ ), info ( $p = .041$ ), and learning ( $p = .003$ ), and writing tasks were significantly higher than decisions ( $p = .003$ ) and learning ( $p = .009$ ).

Assigned task category didn't significantly predict if participants used the AI assistant,  $p = .125$ . AI use was highest in learning tasks and lowest in academic tasks. Tukey pairwise comparisons weren't significant.

```
## Type III Analysis of Variance Table with Satterthwaite's method
##               Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## ai_intent_code 8.8067  4.4034      2   129  38.005 1.056e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##   ai_intent_code emmean      SE  df lower.CL upper.CL
## Asking           0.148 0.0538 129    0.042    0.255
## Doing            0.713 0.0367 129    0.640    0.785
## Expressing       0.389 0.1390 129    0.114    0.664
##
## Degrees-of-freedom method: satterthwaite
## Confidence level used: 0.95

##   contrast      estimate      SE  df t.ratio p.value
## Asking - Doing      -0.564 0.0651 129  -8.659 <0.0001
## Asking - Expressing -0.240 0.1490 129  -1.613  0.2439
## Doing - Expressing   0.324 0.1440 129   2.252  0.0664
##
## Degrees-of-freedom method: satterthwaite
## P value adjustment: tukey method for comparing a family of 3 estimates
```

AI intent significantly predicted Jaccard similarity to the AI response,  $p < .001$ . Responses coded as Doing were much more similar to the AI response ( $M = .713$ ) than responses coded as Asking ( $M = .148$ ). Tukey-adjusted pairwise comparisons showed this difference was significant, Asking - Doing = -0.564,  $p < .001$ .

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: jaccard_similarity_to_ai ~ read_carefully + (1 | task_id)
## Data: subset(reading_depth_data, !is.na(jaccard_similarity_to_ai))
##
```

```
## REML criterion at convergence: 170.6
```

```
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.8733 -0.8683 -0.1436  0.9368  1.6588
##
```

```
## Random effects:
```

```
## Groups Name Variance Std.Dev.
## task_id (Intercept) 0.02479 0.1574
```

```
## Residual 0.15011 0.3874
```

```
## Number of obs: 158, groups: task_id, 18
```

```
##
```

```
## Fixed effects:
```

```
##      Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)  0.62804    0.06096 36.87491  10.303 2.1e-12 ***
## read_carefully -0.19060    0.06476 152.29432  -2.943 0.00376 **
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

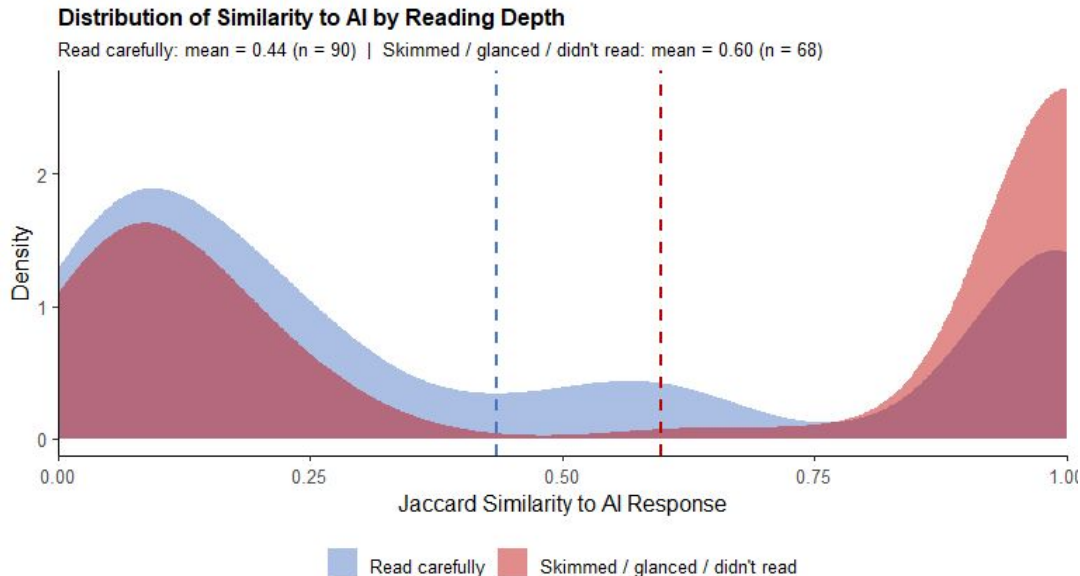
```
##
```

```
## Correlation of Fixed Effects:
```

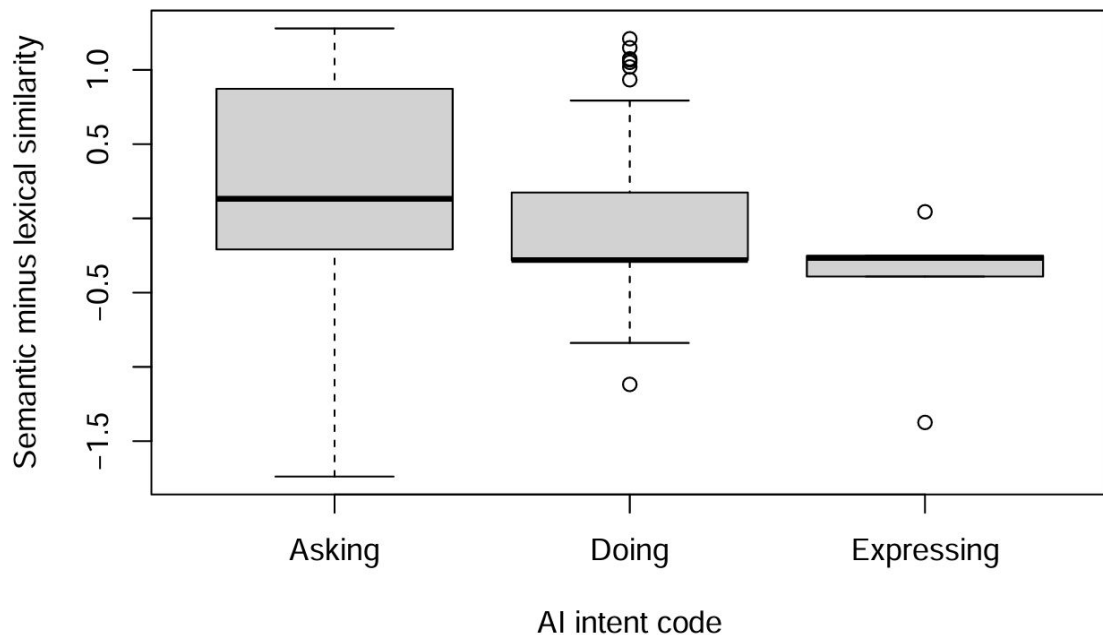
```
##      (Intr)
```

```
## read_crflly -0.606
```

Reading carefully significantly predicted lower Jaccard similarity,  $b = -0.191$ ,  $p = .0038$ .

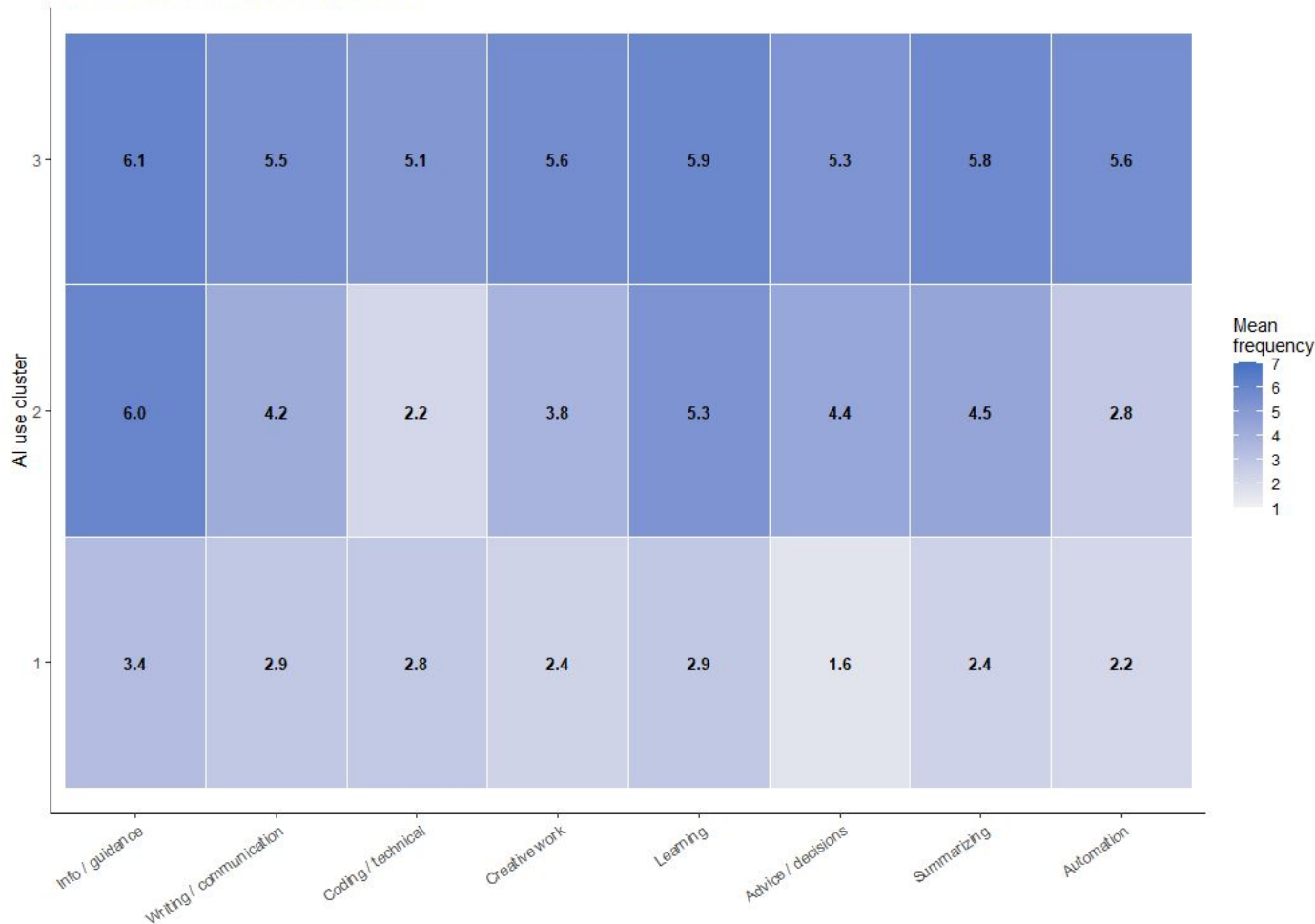


**Divergence by AI intent**



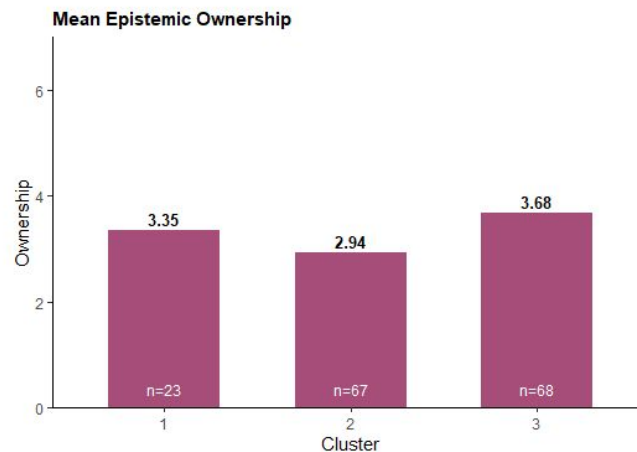
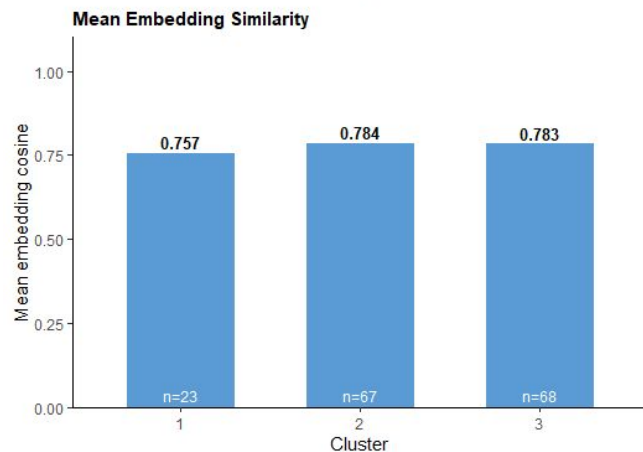
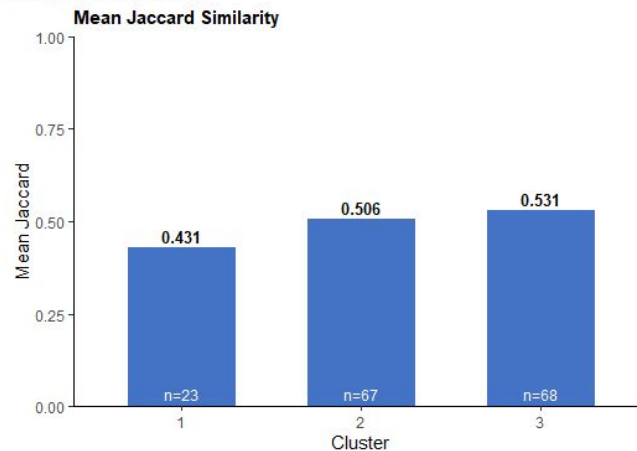
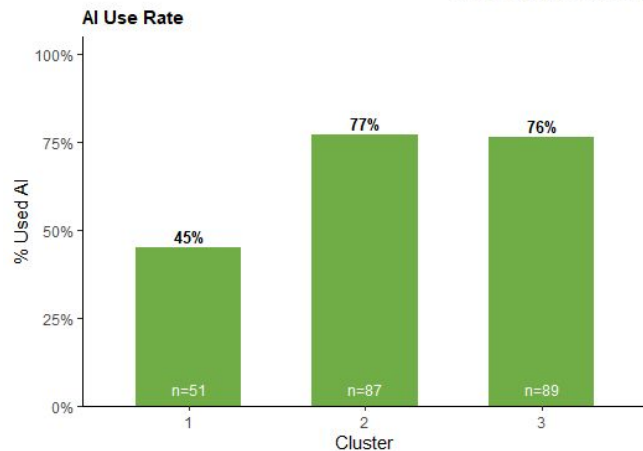
Asking prompts produced the most variable semantic-lexical divergence. Asking responses often had higher semantic similarity relative to lexical overlap, which may indicate idea-level uptake with less direct wording overlap. Doing responses are centered slightly below zero, suggesting somewhat more lexical overlap relative to semantic similarity, but not dramatically so.

AI Use Frequency Profiles by Cluster



- **Cluster 1:** Low-frequency AI users, with relatively low use across all purposes.
- **Cluster 2:** People with relatively high use for information, learning, advice, and summarizing, but lower use for coding, creative work, and automation.
- **Cluster 3:** Broad high-frequency AI users, with high use across all purposes.

## AI Use Clusters and Behavioral Outcomes



Participants' general AI use profiles predicted whether they chose to consult the AI assistant,  $p < .001$ , but not how much they incorporated the AI response once they used it, whether uptake was measured lexically, semantically, or through epistemic ownership.